Fiscal Year:	FY 2016	**Task Last Updated:**	FY 11/12/2015
PI Name:	Turker, Mitchell Ph.D.		
Project Title:	The Relation Between Mutagenesis and Genomic Instability After Particle Exposure In Vivo		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	97239-3011	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Task Description:	Astronauts in the space environment are at risk for a variety of health-related concerns from exposure to accelerated ions (fast moving atoms that have lost their electrons) that can pass through the spacecraft hull and interact with astronauts' cells within the body. These particles are released from the sun in solar particle events (SPE) and exist as part of the Galactic Cosmic Ray (GCR) background radiation, although the types of ions and their energies can differ widely in SPE and GCR. A significant concern for astronaut health is an increased risk of cancer, which is linked to a variety of radiation exposures. The underlying cause of cancer, regardless of its source, is mutations in critical genes and ongoing gains, losses, and rearrangement of chromosomes. Collectively, these ongoing changes in the cell are termed genomic instability. The goal of the proposed work is to determine how frequently different types of accelerated ions found in the GCR cause mutations and genomic instability and to describe what these changes look like. To better extrapolate our results to potential harms to astronauts from space radiation, we will use a mouse model for the proposed work. This will ensure that the cell types of interest found in the spleen and kidney will be exposed to accelerated ions in their normal tissue environment. We will focus on accelerated ions found in space for which relatively little information is currently available. Successful completion of the proposed work will help in the design of long-term travel for humans in space and aid in the protection of space travelers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are currently developing a new cytogenetic based method to identify radiation-induced mutations directly in cell nuclei. If we are successful, this method will apply to a number of research projects including those that do not involve radiation exposure.
Task Progress:	1. Mouse irradiation with calcium ions. We are using a mouse strain designed to detect mutations that form in the body to measure the frequency and types of mutations caused by accelerated charged particles (ions) found in space. Accelerated charged particles are atoms that are stripped of electrons, which is why they are positively charged. On June 22, 2015 we irradiated 72 mice with accelerated calcium (Ca) ions; 24 each at 3 different doses that correspond to approximately 1, 2, or 8 particles passing through each cell nucleus, respectively. We also tested 18 mice that were not irradiated, as controls. We are currently setting up cell cultures to isolate mutant cells from mouse kidneys and spleens, which represent two very different cell types. Cells from the kidney are termed epithelial cells, and such cells make up most of the organs found in the body and give rise to most cancers in humans. The spleen contains T cells, which are a type of white blood cell the body uses to fight infections. For each mouse, one kidney and the spleen are removed 3-5 months after exposure, cell suspensions are made, and the cells plated in tissue culture dishes to quantify mutant frequencies and determine cell viability. The cell viability data will be used to determine relative levels of lingering toxicity in the cells of each tissue as a result of exposure to the different doses of Ca ions. The mutant frequency data will be used to determine the shape of the dose response curve for the induction of mutants. DNA from the mutant cells will be used to determine the types of mutations induced by the Ca ions at different doses. The mice ranged in age from 5-7 months at the time of irradiation and are evenly divided between males and females. This process was used previously to isolate and characterize all mutant cells discussed below. 2. T cell mutants selected from silicon (Si) ion-irradiated mice. In June 2014, 99 mice were exposed to Si ions, again at doses corresponding to 1, 2, or 8 particles per cell nucleus. We started isolating mutant spleen T cells and kidney epithelial cells in fall 2014. The T cell mutants have been collected and are awaiting molecular analysis in year 3. The kidney cultures, which take many more weeks in the incubator (5-6 weeks), had frequent fungal contamination. This is an issue that crops up in older incubators (>10 years) for long-term cultures. We lost most of these cultures and will not get meaningful data for the kidneys (though fortunately we have data from mouse kidneys exposed to somewhat higher doses of Si ions, see below). In response, and to prepare for the calcium ion irradiated kidney cultures (discussed above in #1), we were able to obtain two new NuAire cell culture incubators with assistance from the Vice President's Office for Research at Oregon Health & Science University (OHSU). A unique feature of these incubators is that they can be sterilized after each set of kidney cultures preps, which eliminates the fungal contamination problems in the future. 3. Completed molecular and statistical analysis of mutations in spleen T cell mutants from mice exposed to titanium (Ti) ions, iron (Fe) ions, protons, or sham-irradiated (i.e., no exposure). In preparation for a publication on charged particle mutagenesis in mouse spleen T cells, we finished the molecular and statistical analysis for mutants isolated from spleens of mice exposed to Ti ions, Fe ions, and protons. For comparison, we isolated mutant cells from non-irradiated mice to identify the types of mutations that occur spontaneously in the mouse spleen. The doses used for Fe ions and Ti also corresponded to 1, 2, and 8 particles per nucleus. Mutant frequencies in the control non-irradiated mice were approximately 1 to 3 per 100,000 spleen T cells. In comparison, the spontaneous mutant frequency in the kidney epithelial cells is approximately 10 per 100,000 cells. Surprisingly, mutant frequencies in the irradiated mice did not increase relative to the controls for any of the particles or doses tested. This is in contrast to the kidneys in the same mice where mutant frequencies rose notably after exposures to Fe ions and Ti ions. Despite the lack of increased mutant frequencies in the irradiated spleens, a molecular analysis revealed that the exposures induced radiation signature mutations (losses of large pieces of a target chromosome), with the highest levels of induction at 2 particles of Fe ions and Ti ions per nucleus. These results show that charged particles found in the space environment can induce mutations even if a T cell is struck by only a small number of charged particles. 4. Continued development of new method to detect radiation signature mutations in intact cells. One goal of current funding is to create a mutation assay to detect radiation signature mutations in a relatively small number of cells (~50). The molecular assay discussed above require DNA that is isolated from approximately 100,000 cells. The focus of the new assay is to identify mutations that remove significant regions of the target chromosome because we know from molecular and cytogenetic work that these mutations are far more common in charged particle-induced mutant cells than in spontaneous mutants. During the last year we completed establishing this method by first examining four Fe ion-induced kidney mutant clones that we had previously characterized to have mutations removing a significant fraction of the target chromosome. The new assay easily detected the loss of this genetic material. We next examined 8 new mutant kidney clones from mice exposed to oxygen (O) or silicon (Si) ions that had not been characterized previously, and the assay detected radiation signature mutations in 3 of them. A molecular assay confirmed these observations. 5. Completed molecular analyses for Aprt kidney mutants isolated from mice exposed to Fe ions, Si ions, or O ions. In addition to the spleen T cell mutants that were characterized during the last year (#3, above), we have finished the molecular characterization of mutant kidney epithelial cells from from mice exposed to Fe ions, Si ions, and O ions The molecular characterization included the target chromosome and additional non-selected chromosomes with a focus on

	chromosome 14, which we have shown to be preferentially affected in the charged particle-induced mutants. These data will be analyzed statistically during the next year and submitted for publication. 6. Plans for Year 3. Our plans for the next year are: 1) to complete the isolation and then characterization of kidney epithelial cell and spleen T cell mutants from mice exposed to Ca ions (See #1 above), 2) characterize T cell mutants isolated during years 1 and 2 from mice exposed to Si and O ions, and 3) prepare manuscripts for publication. Manuscripts that will be written for which data is complete or nearly so are for: 1) spleen T cell mutants induced by Fe ions, Ti ions, and protons (see #2 above); 2) analysis of mutations in kidney cells induced by 3 doses of Fe ions; 3) analysis of mutations in kidney cells induced by 3 doses of O and Si ions; and 4) description of the new assay to characterize radiation-induced mutations (#4 above).
Bibliography Type:	Description: (Last Updated: 11/01/2023)
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